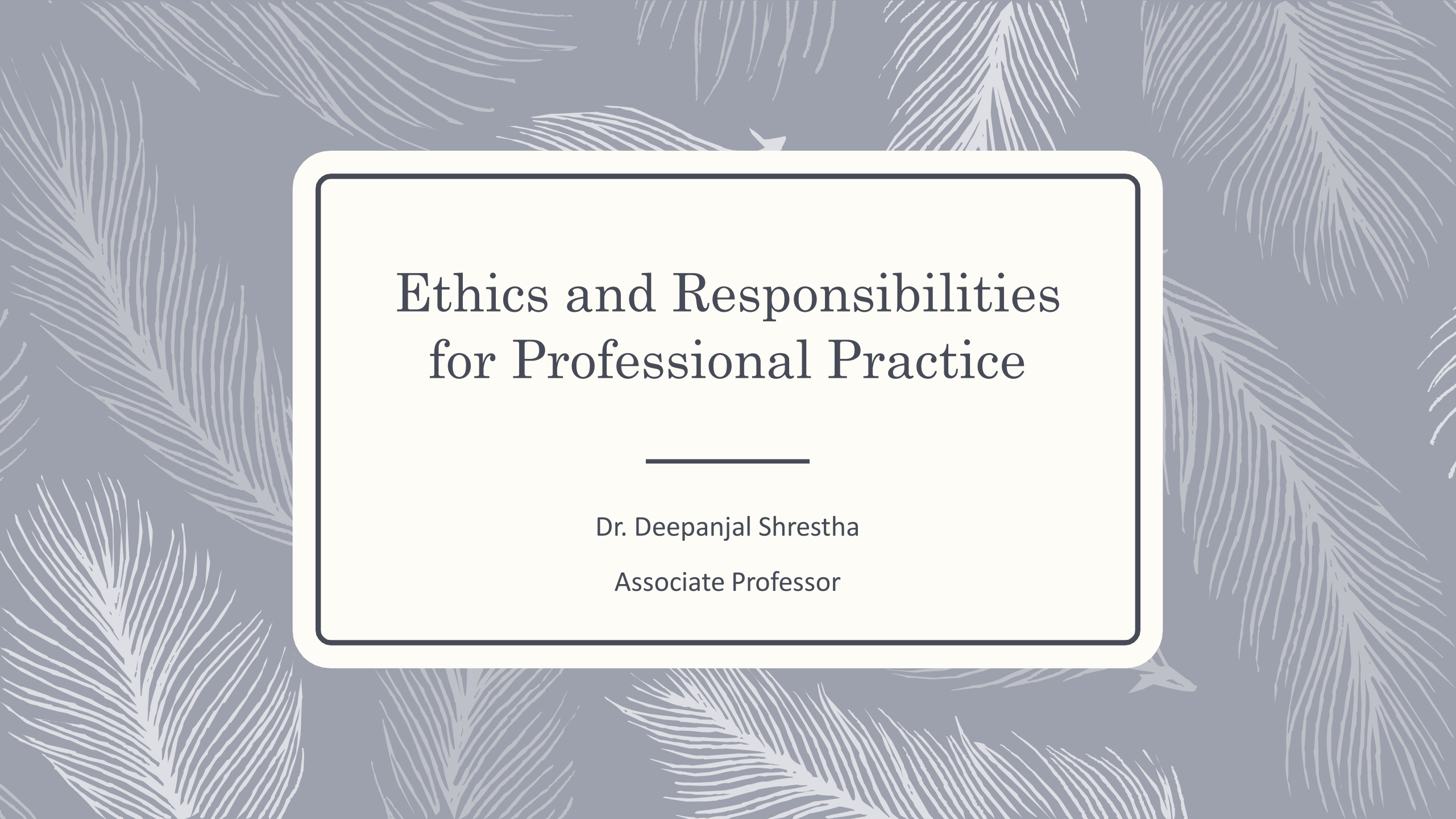




Ethics and Responsibilities for Professional Practice

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Introduction to Professional Ethics and Responsibility

- Professional ethics refers to the moral principles and standards that guide the behavior of professionals in their work.
- Responsibility means being accountable for the consequences of one's professional actions on society, clients, employers, and the profession itself.



Roles of a professional

- Problem solver using specialized knowledge
- Advisor to clients or organizations
- Guardian of safety, quality, and reliability
- Representative of the profession to society



Core responsibilities

- Competence: work only within one's expertise
- Honesty: truthful reporting and communication
- Integrity: avoiding deceptive practices
- Accountability: accepting responsibility for outcomes
- Respect for society: considering social and environmental impact



Profession

- A profession is an occupation that requires
 - extensive training
 - extensive study
 - mastery of specialized knowledge, usually has a professional association,
 - ethical code and
 - process of certification or licensing.



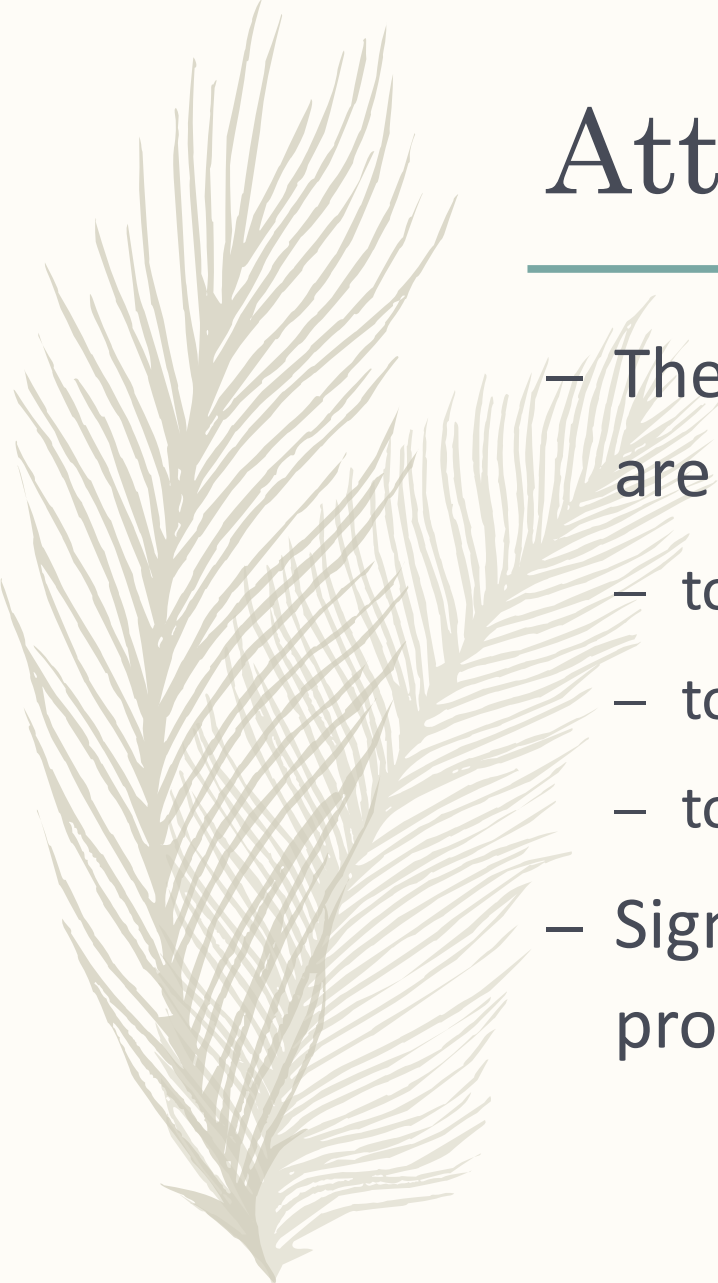
Examples of Profession

- Examples are:
 - librarianship,
 - accounting,
 - law,
 - teaching,
 - architecture,
 - medicine,
 - military,
 - the clergy, nursing,
 - those who work or perform research in the various sciences, or engineering.



Attributes of Profession

- Work that requires sophisticated skills,
 - the use of judgment, and the exercise of discretion.
 - Also, the work is not routine and is not capable of being mechanized;
- Membership in the profession requires
 - extensive formal education,
 - not simply practical training or apprenticeship;



Attributes

- The public allows special societies or organizations that are controlled by members of the profession
 - to set standards for admission to the profession,
 - to set standards of conduct for members, and
 - to enforce these standards; and
- Significant public good results from the practice of the profession.

Examination of Different Occupations

- Athletes in light of the foregoing definition of professions
 - Athletics requires extensive training, not of a formal nature, but more of a practical nature acquired through practice and coaching.
 - No special societies are required by athletes, and athletics does not meet an important public need;
 - although entertainment is a public need, it certainly doesn't rank highly compared to the needs met by professions such as medicine.
 - So, although they are highly trained and very well compensated, athletes are not professionals





Example 2

- Medicine certainly fits the definition of a profession given previously.
- It requires very sophisticated skills that cannot be mechanized,
- It requires judgment as to appropriate treatment plans for individual patients, and
- it requires discretion



Further

- Medicine has a special society
 - (e.g. Nepal Medical Council)
- To which a large fraction of practicing physicians belong and that participates in the regulation of medical schools, sets standards for practice of the profession, and
- Enforces codes of ethical behaviour for its members



Is Engineering a Profession ?

- **Engineering is a profession because: -**
 - It needs extensive and formal education / training
 - Engineering design requires sophisticated skills that can not be mechanized. (Only the construction after the design could be mechanized.)
 - Engineering design involves a remarkable amount of judgement and discretion.
 - Engineers have societies (e.g. IEEE, ACM etc.) which set standards for engineering practice and code of ethics for the professionals.
 - They produce publicly important results like bridges, automated machines etc.



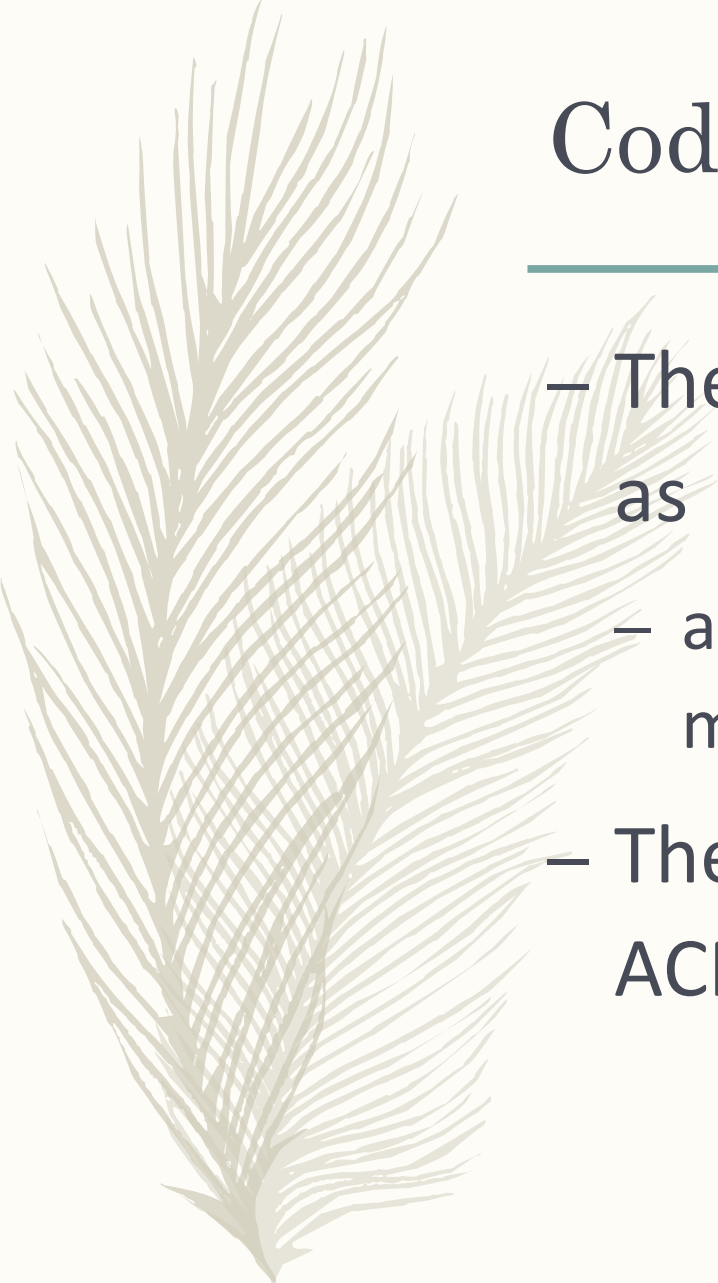
Professional Responsibilities

- Confidentiality and Proprietary Information
 - keep certain information of the client secret or confidential
- Conflict of Interest
 - Avoiding conflict of interest is important in any profession
- Environmental Ethics
 - to protect the integrity of the biosphere
 - to ensure a healthy environment for humans



Professional Rights

- The most fundamental right is professional conscience
 - Right to exercise professional judgement in discharging one's duties and to exercise this judgement in an ethical manner
- Rights include
 - *The right to privacy,*
 - *The right to participate in activities of one's own choosing outside of work,*
 - *The right to reasonably object to company policies without fear of retribution, and the right to due process*



Codes of Ethics and Professional Conduct

- The most important function of a code of ethics is as
 - a statement embodying the collective wisdom of members of the profession
- The most visible codes are those produced by the ACM and the IEEE



The Institute of Electrical and Electronics Engineers, Inc. (IEEE)

- To accept responsibility in making engineering decisions consistent with the safety, health and welfare of the public, and to disclose promptly factors that might endanger the public or the environment;
- To avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
- To be honest and realistic in stating claims or estimates based on available data;
- To reject bribery in all its forms
- Etc...



ACM Code of Ethics and Professional Conduct

- Contribute to society and human well-being
- Avoid harm to others
- Be honest and trustworthy
- Be fair and take action not to discriminate
- Honour property rights including copyrights and patent
- Give proper credit for intellectual property
- ETC...



Knock your mind!

- Example

A software engineer knowingly releases insecure software to meet deadlines.

- Who is responsible if users suffer losses?

List one responsibility that you think is most important for your future profession and explain why.



Conflict of Interest

- A conflict of interest occurs when a professional's personal interest interferes with their duty to clients, employer, or public.
- Types of conflict
 - Financial conflict
 - Personal relationship conflict
 - Organizational conflict



Examples: Conflict of Interest

- An engineer approving a project owned by a relative
- A consultant recommending a product from a company they secretly own shares in



Whistleblowing

- Whistleblowing is reporting unethical, illegal, or harmful activities within an organization to authorities or the public.
- When whistleblowing is ethically justified
- Serious harm to public or users
- Internal reporting mechanisms have failed
- Evidence is credible
- Intent is to prevent harm, not personal gain



Risks involved

- Job loss
- Legal action
- Social pressure
- Example
 - A bank employee reports massive data privacy violations ignored by management.
- Discussion question
 - Is loyalty to employer more important than responsibility to society?



Corporate Social Responsibility

- Explain CSR as ethical obligation beyond profit.
- CSR includes
 - • Ethical business practices
 - • Environmental sustainability
 - • Fair treatment of employees
 - • Contribution to community development



Technology related CSR examples

- Data privacy protection
- Responsible use of AI
- Reducing digital divide

Nepali context example

- IT companies supporting digital literacy programs in rural Nepal.



Intellectual Property

- Property that can be protected under federal law, including copyrightable works, ideas, discoveries, and inventions is known as Intellectual property.
- In law, particularly in common law jurisdictions, intellectual property or IP refers to a legal entitlement which sometimes attaches to the expressed form of an idea, or to some other intangible subject matter.



Types of Intellectual property

- Well known forms of intellectual property include copyrights, patents, trademarks, and trade secrets.
- Patents and trademarks fall into a particular subset of intellectual property known as industrial property.



Copyright

- Copyright is a protection that covers published and unpublished literary, scientific and artistic works, whatever the form of expression, provided such works are fixed in a tangible or material form.
- Copyright laws grant the creator the exclusive right to reproduce, prepare derivative works, distribute, perform and display the work publicly.



Copy right Start / End

- Copyright protection begins when any of the above described work is actually created and fixed in a tangible form.



Example

- Assume a person is a musician, when he writes new lyrics, he prints them out on paper, signs his name at the bottom with the Copyright © symbol to show that he is the author, places it in an envelope and mails it to himself without opening it.
- His copyright begins at the moment he puts his idea in a tangible form by printing the lyrics out on paper.
- He creates proof when he mails it to himself - the postmark establishes the date of creation.



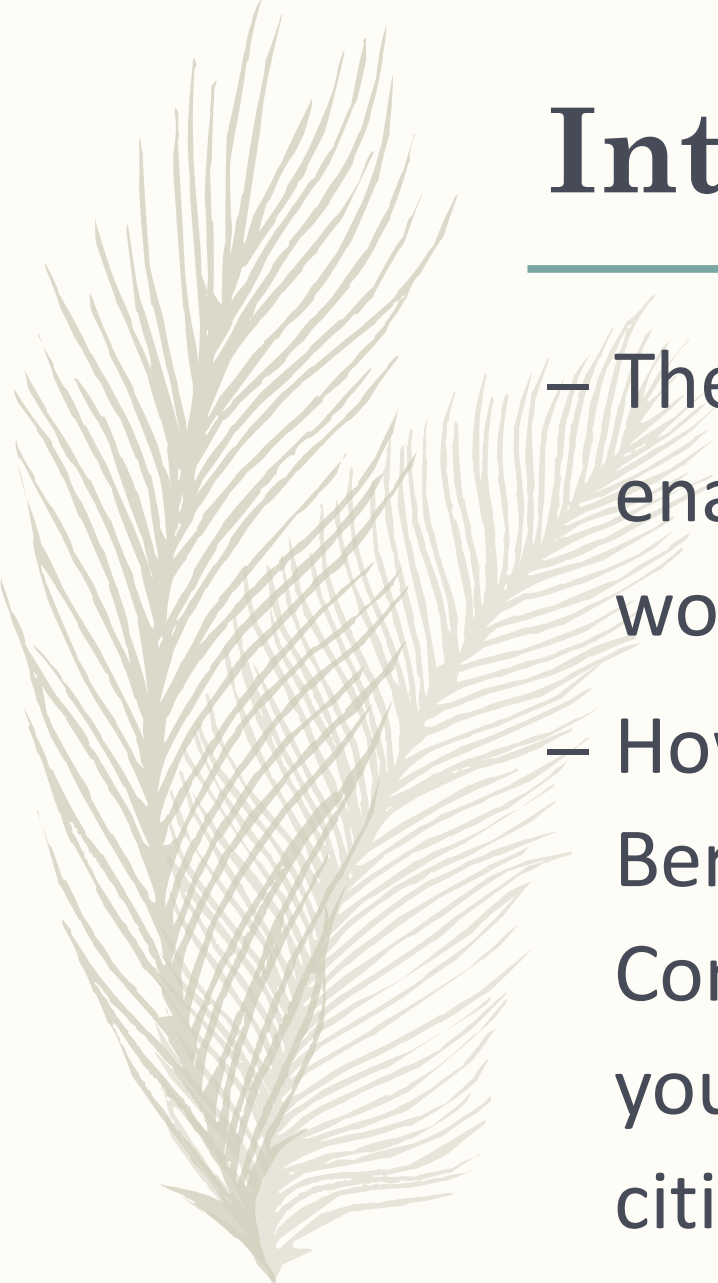
Contd..

- He then registers his copyright with the legal body (like U.S. Copyright Office in US) which is a requirement in order to sue for monetary damages should a violation of his copyright arise.
- However, if somebody copies and redistributes his lyrics without permission before his copyright is registered, he still has the right to assert a copyright claim as the true author.



Expiration of Copyright

- The Berne Convention establishes a general and minimum period that lasts the life of the author and fifty years after his (or her) death.
- Cinematographic works and photographic works have a minimum period of protection of 50 and 25 years upon the date of creation, respectively.



International Copyrights.

- There are no "international copyrights" that enable you protect your work throughout the world.
- However, most countries are members of the Berne Convention and the Universal Copyright Convention (UCC), which allow you to protect your works in countries of which you are not a citizen or national



Patent

- Patent is any new method or way of constructing, conducting or processing any matter or body of matters or any useful invention based on a new principle or formula
- Patents must be registered
- Infringement of patent laws is subject to fine of up to RS. 2,000 and confiscation of articles connected with the offence



Design

- Design means any feature, pattern or shape of a matter prepared and produced in any manner.
- Designs must be registered
- Infringement of design law is subject to fine upto Rs. 800 and confiscation of articles connected with the offence.



Trademark

- Trademark is any sign, picture or word or their combination used by a person, firm or company to distinguish its goods and services from those of others.
- A registered brand is a trademark.
- Registration is needed to obtain right of trademark.
- The term of registration is seven years.



Comparison with patents designs and copyright

- While trademark law seeks to protect indications of the commercial source of products or services,
- Patent law generally seeks to protect new and useful inventions, and registered designs law generally seeks to protect the look or appearance of a manufactured article.
- Trademarks, patents and designs collectively form a subset of **intellectual property** known as industrial property.



Trade secret

- A trade secret is a confidential practice, method, process, design, or other information used by a company to compete with other businesses.
- It is also referred to in some jurisdictions as confidential information.



Trade secret contd..

- The concept is that, sometimes, Company A is more successful than Company B, or is successful at all, not so much due to access to markets, resources, or personnel, but due to special knowledge owned by Company A.
- If others had access to the same knowledge, then Company A's ability to survive in an otherwise equal marketplace would be impaired.
- Thus, such secrets are guarded zealously.



ICT Policy of Nepal

- The ICT Policy of Nepal provides a national vision for the development, regulation, and promotion of information and communication technologies.
- Objectives of ICT Policy
 - Promote digital transformation and e governance
 - Encourage ICT based innovation and entrepreneurship
 - Expand ICT infrastructure across the country
 - Develop skilled human resources in ICT
 - Support private sector participation in IT industry



Key focus areas

- • Digital Nepal Framework
- • Broadband and internet expansion
- • Software and IT service exports
- • Cyber security and data protection
- • Use of ICT in education, health, and governance

Relevance for professionals

- • Guides national IT development priorities
- • Encourages startups and innovation
- • Provides policy backing for IT investments



Electronic Transactions Act (ETA), Nepal

- The Electronic Transactions Act provides legal recognition to electronic records and digital transactions.
- Major provisions
 - • Legal validity of electronic records
 - • Recognition of digital signatures
 - • Regulation of electronic contracts
 - • Cyber crime offenses and penalties



Cyber crimes covered under ETA

- • Unauthorized access to computer systems
- • Data theft and manipulation
- • Online fraud and identity theft
- • Publication of illegal digital content



Importance for IT professionals

- Software developers must ensure secure systems
- E commerce platforms must follow digital contract rules
- Professionals can be legally liable for cyber offenses

Example

- Online banking, digital payments, and e commerce transactions are legally valid under this Act.



Communication Act, Nepal

- The Communication Act regulates communication services and media in Nepal.

Key objectives

- Regulation of telecommunications services
- Licensing of communication service providers
- Management of radio frequencies
- Ensuring fair access to communication services



Communication Act, Nepal

Scope

- Telecom operators
- Internet service providers
- Broadcasting services
- Digital communication platforms

Relevance to IT sector

- ISPs and telecom based startups must follow licensing rules
- Communication services must comply with regulatory standards
- Misuse of communication systems can result in legal action



Legal Issues for Startup in Nepal

- Legal requirements for starting and operating a business in Nepal, especially technology startups.

Business Registration in Nepal

- Common forms of business
 - • Sole proprietorship
 - • Partnership firm
 - • Private limited company



Legal Issues for Startup in Nepal

- Basic registration steps
 - Register business with Office of Company Registrar or local authority
 - Obtain Permanent Account Number (PAN)
 - Register for Value Added Tax (VAT) if applicable
 - Register with local municipality or ward office

Importance of registration

- Legal identity of business
- Ability to open bank accounts
- Eligibility for loans and investment
- Compliance with government regulations



Taxation and Compliance Requirements

Major taxes applicable to startups

- Income tax on profit
- Value Added Tax (VAT) if turnover exceeds threshold
- Withholding tax on salaries and services

Compliance requirements

- Regular tax filing
- Annual financial statements
- Audit requirements for companies
- Renewal of licenses if applicable



Why compliance matters?

- Avoid fines and legal penalties
- Build credibility with investors and customers
- Enable long term business sustainability



Conclusion and Summary

Thank you